

BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

Complete

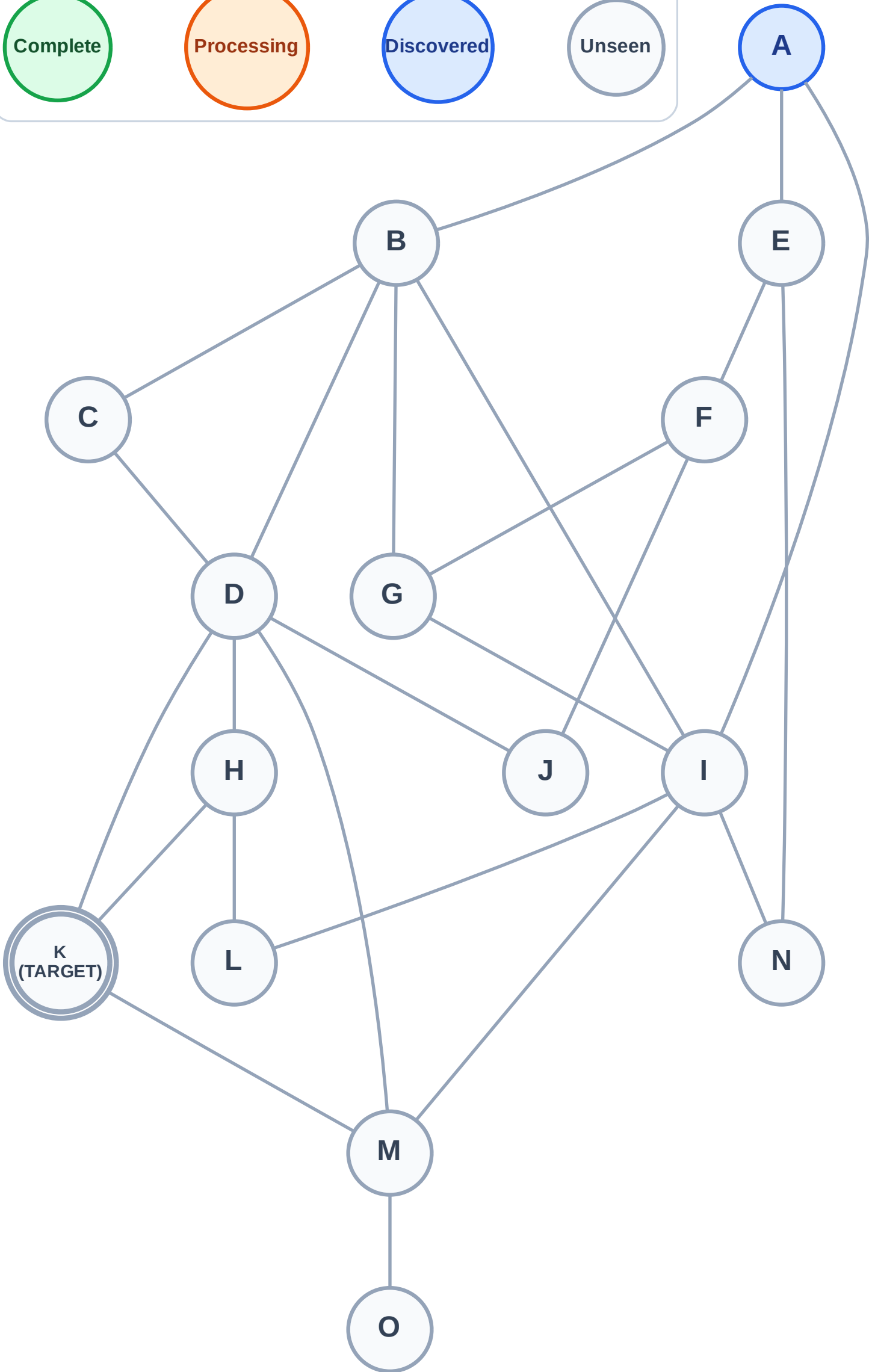
Processing

Discovered

Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

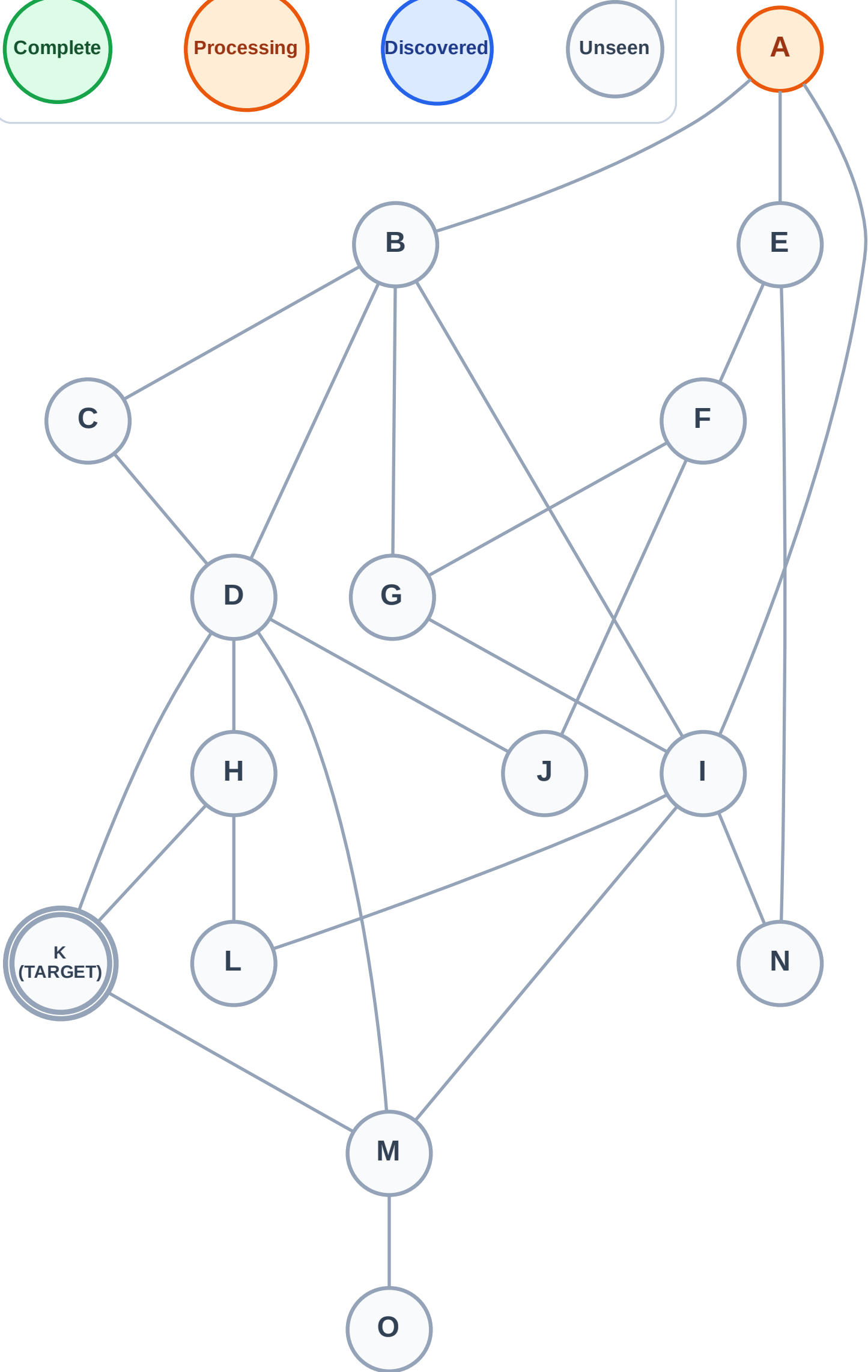
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

Step 3: Discover B, E, I from A

Legend

Complete

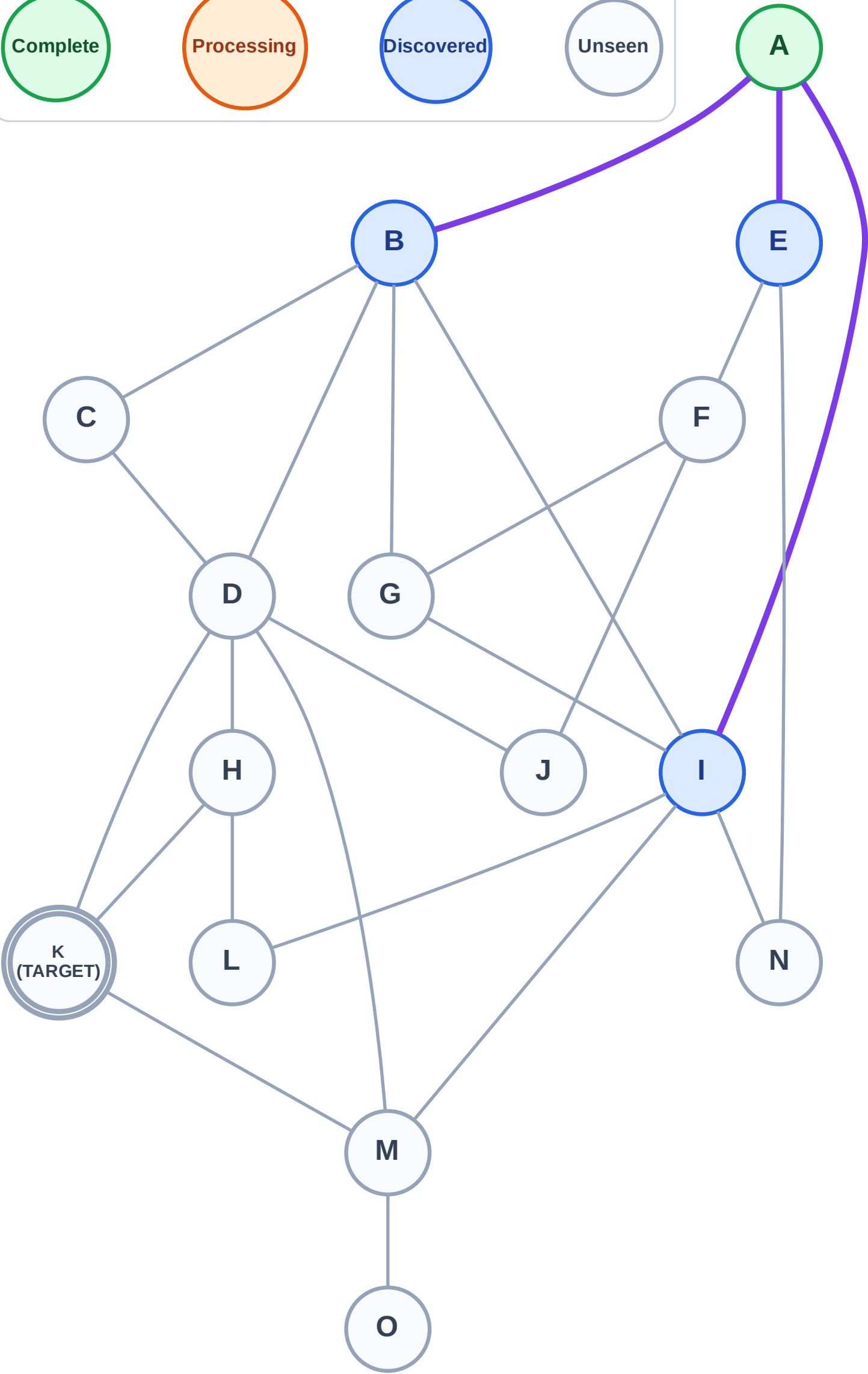
Processing

Discovered

Unseen

Queue: B → E → I

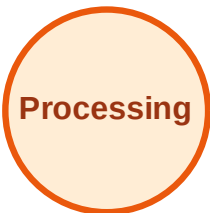
Visited: A



BFS Traversal

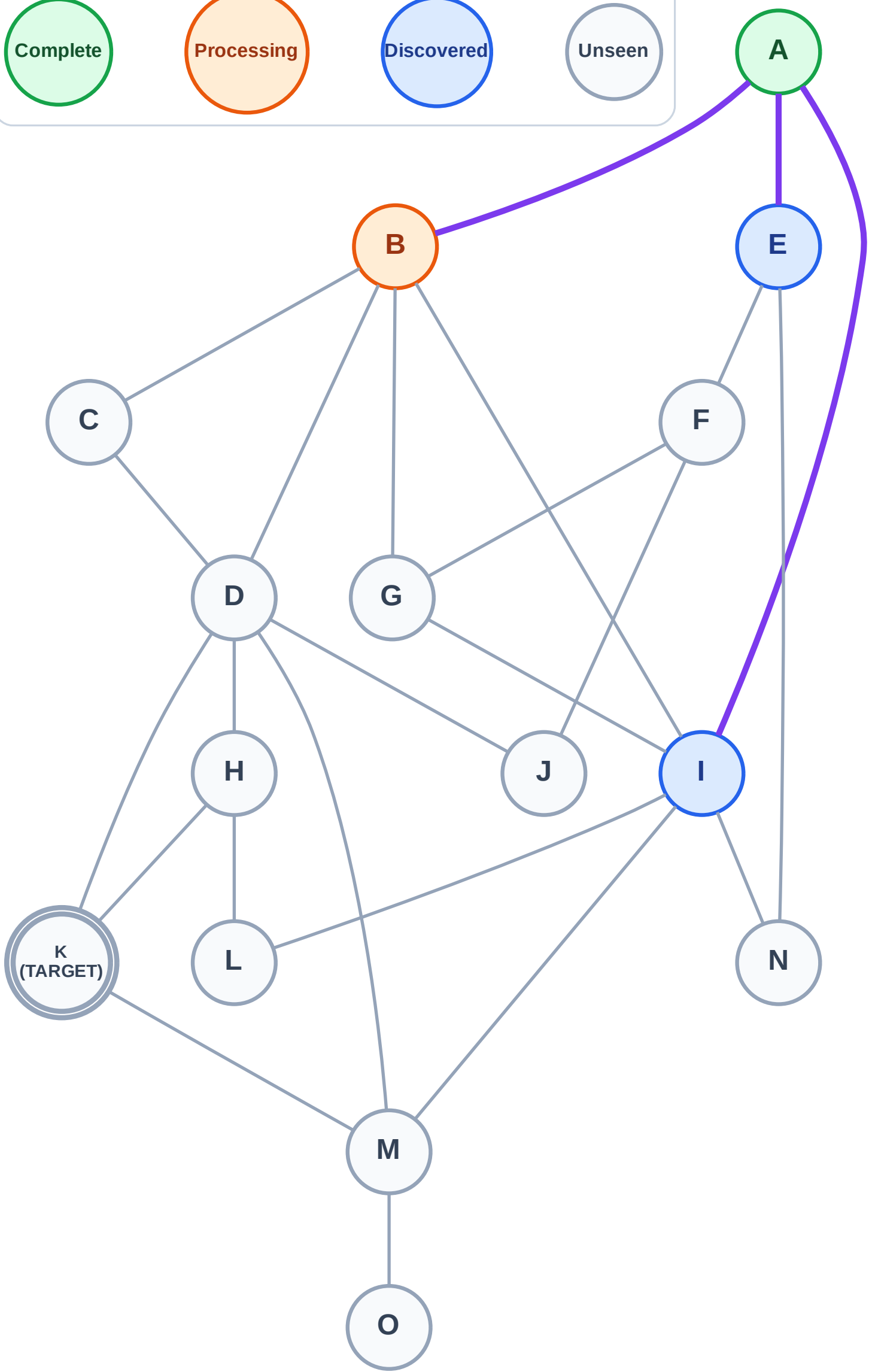
Step 4: Process node B

Legend



Queue: E → I

Visited: A



BFS Traversal

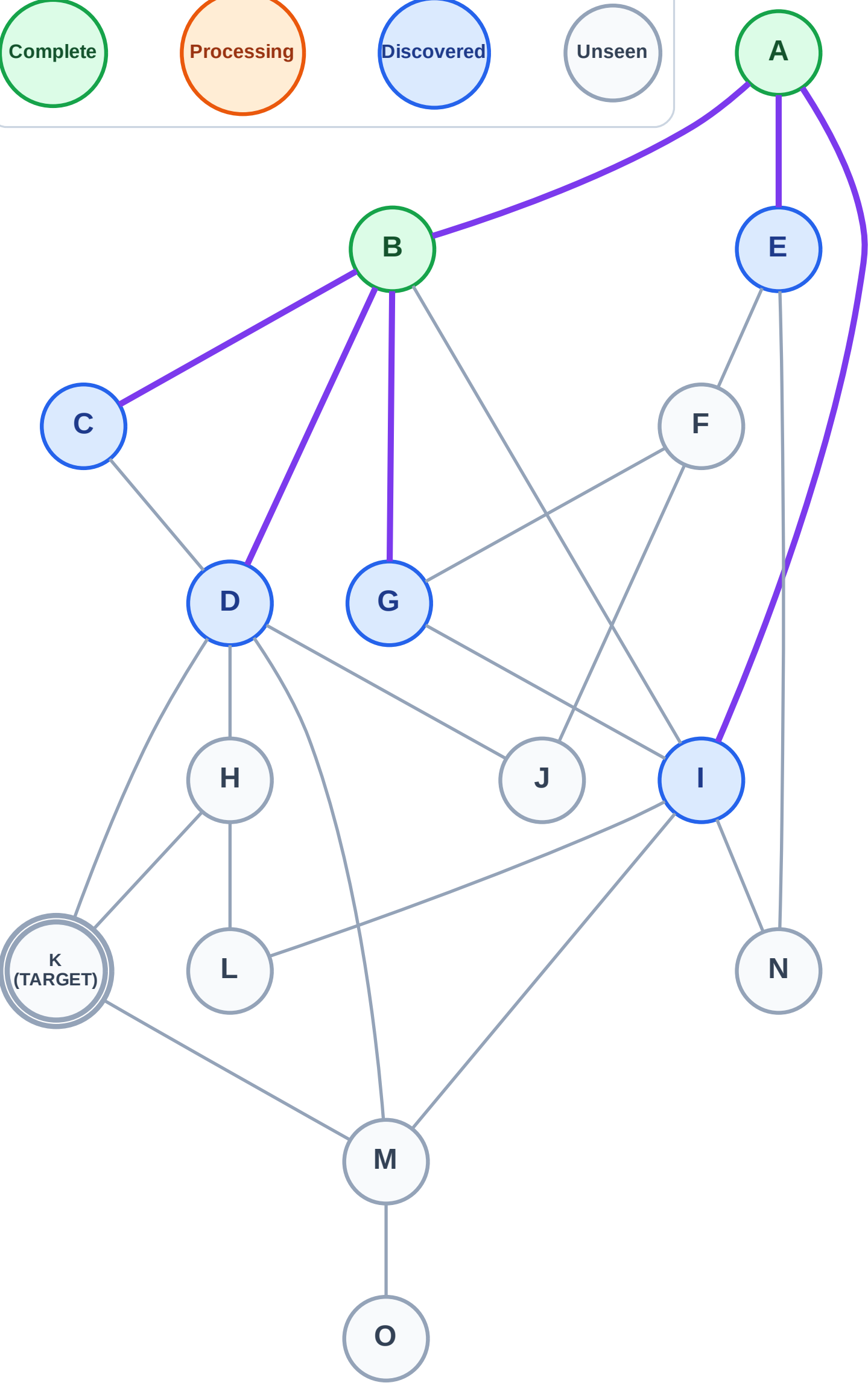
Step 5: Discover C, D, G from B

Legend



Queue: E → I → C → D → G

Visited: A, B



BFS Traversal

Step 6: Process node E

Legend

Complete

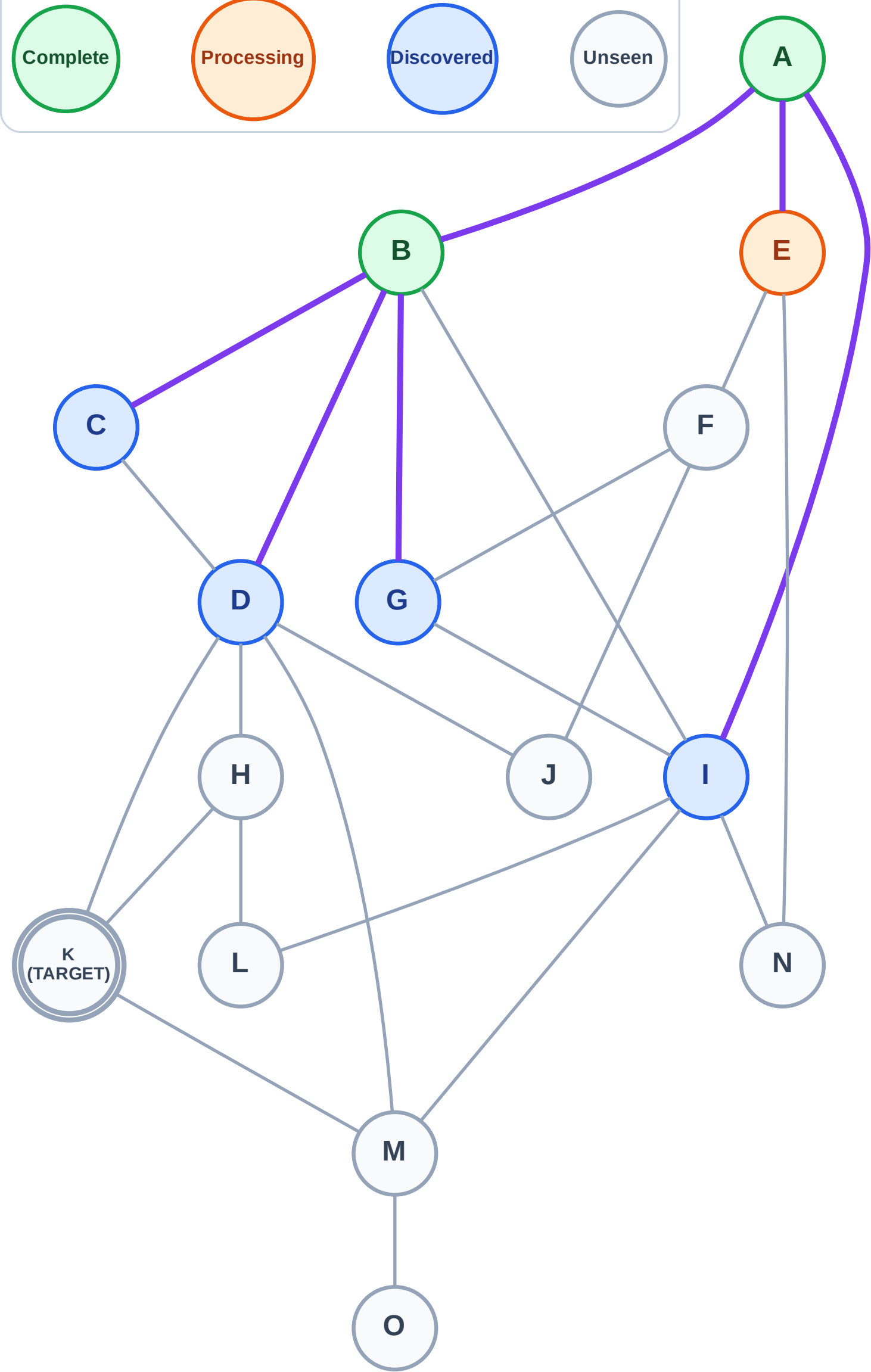
Processing

Discovered

Unseen

Queue: I → C → D → G

Visited: A, B



BFS Traversal

Step 7: Discover F, N from E

Legend



Complete



Processing



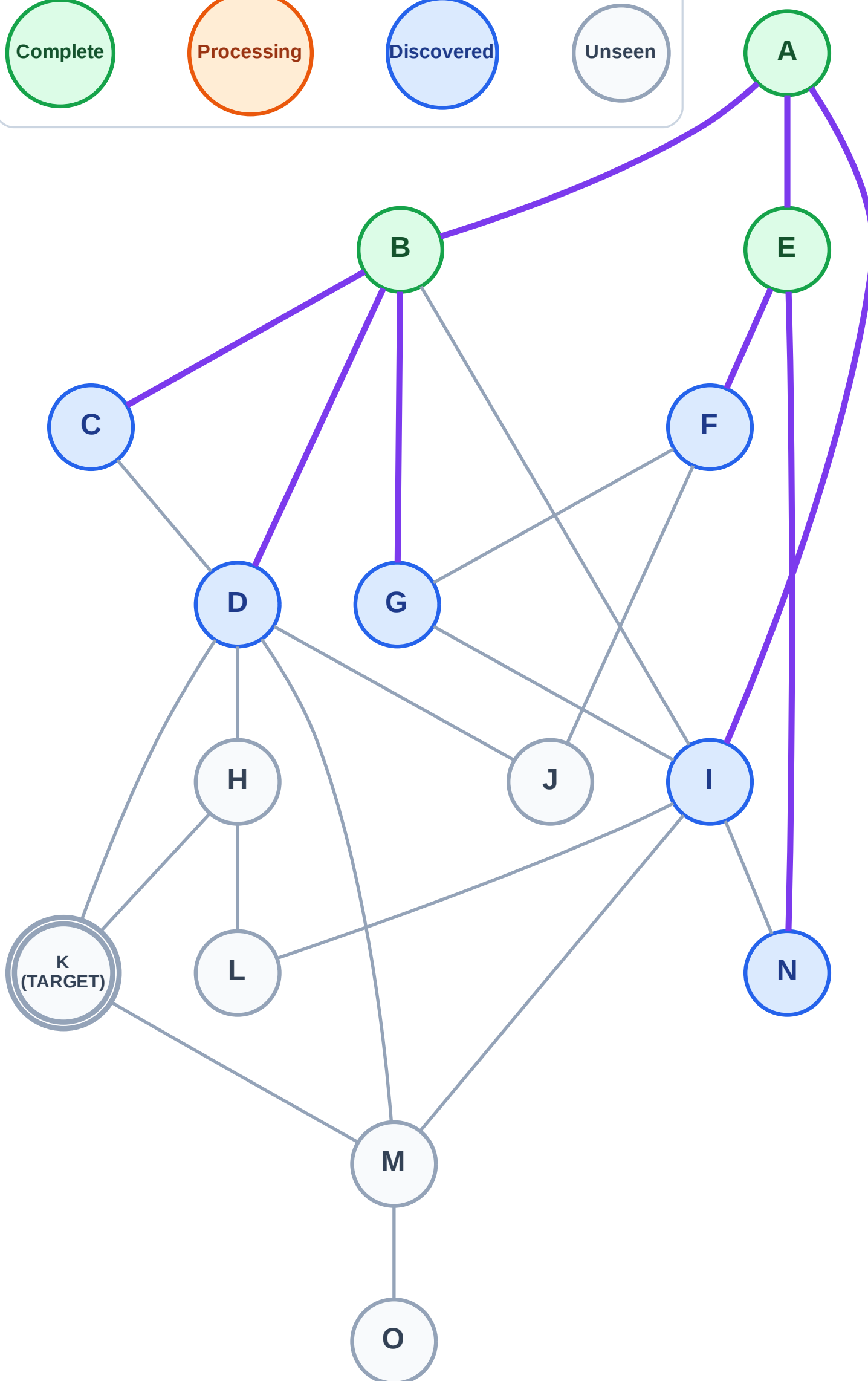
Discovered



Unseen

Queue: I → C → D → G → F → N

Visited: A, B, E



BFS Traversal

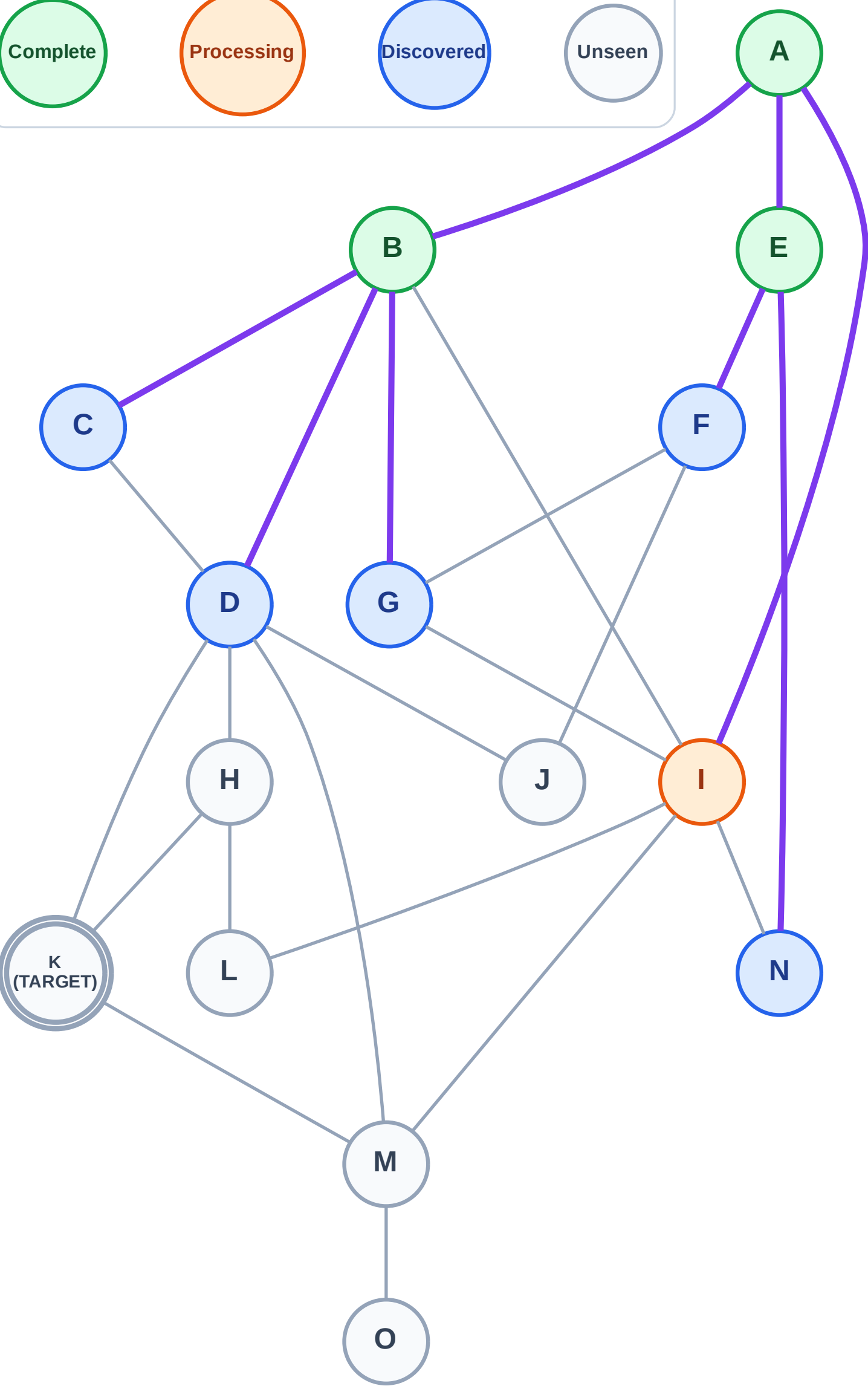
Step 8: Process node I

Legend



Queue: C → D → G → F → N

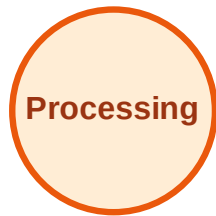
Visited: A, B, E



BFS Traversal

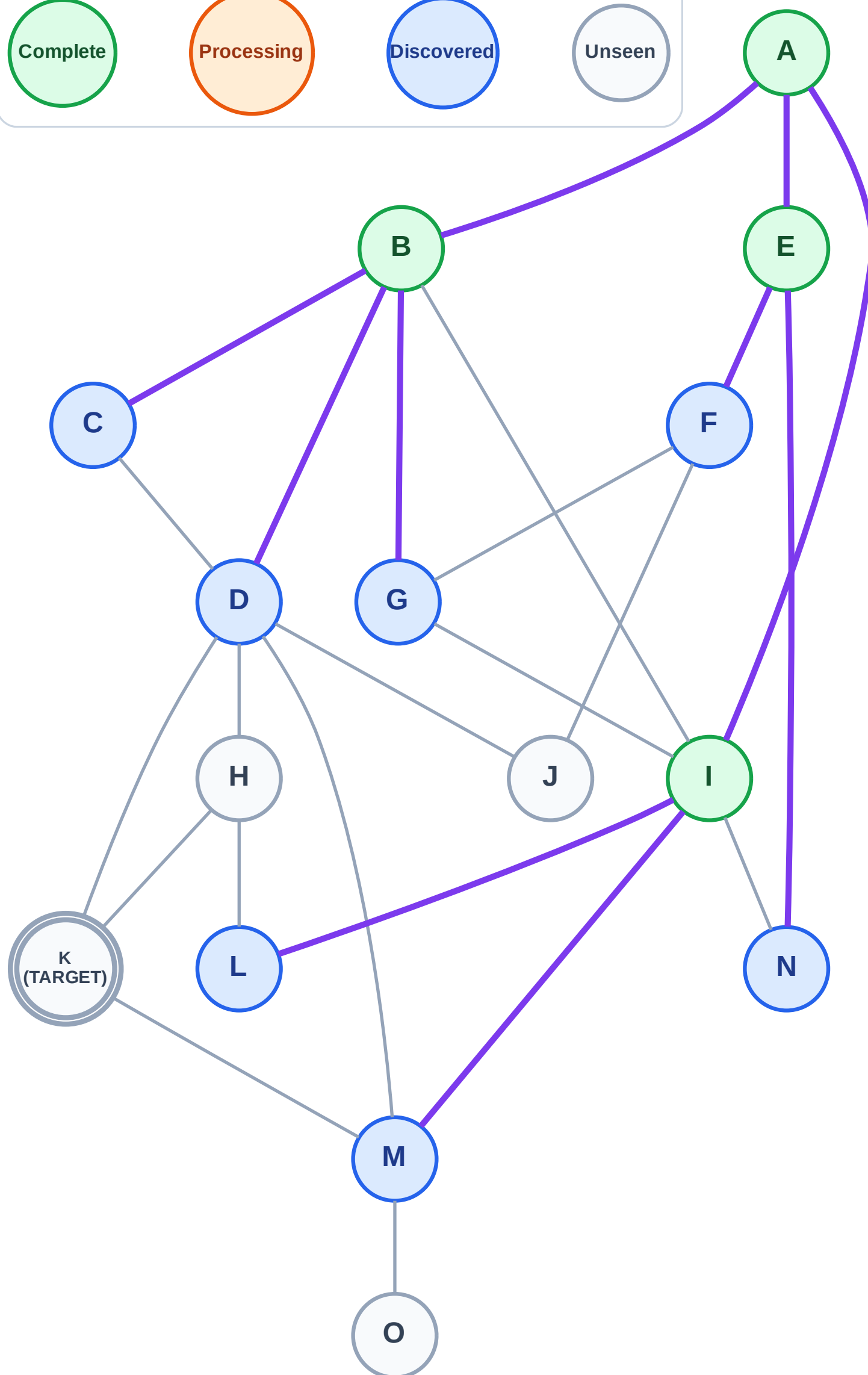
Step 9: Discover L, M from I

Legend



Queue: C → D → G → F → N → L → M

Visited: A, B, E, I



BFS Traversal

Step 10: Process node C

Legend

Complete

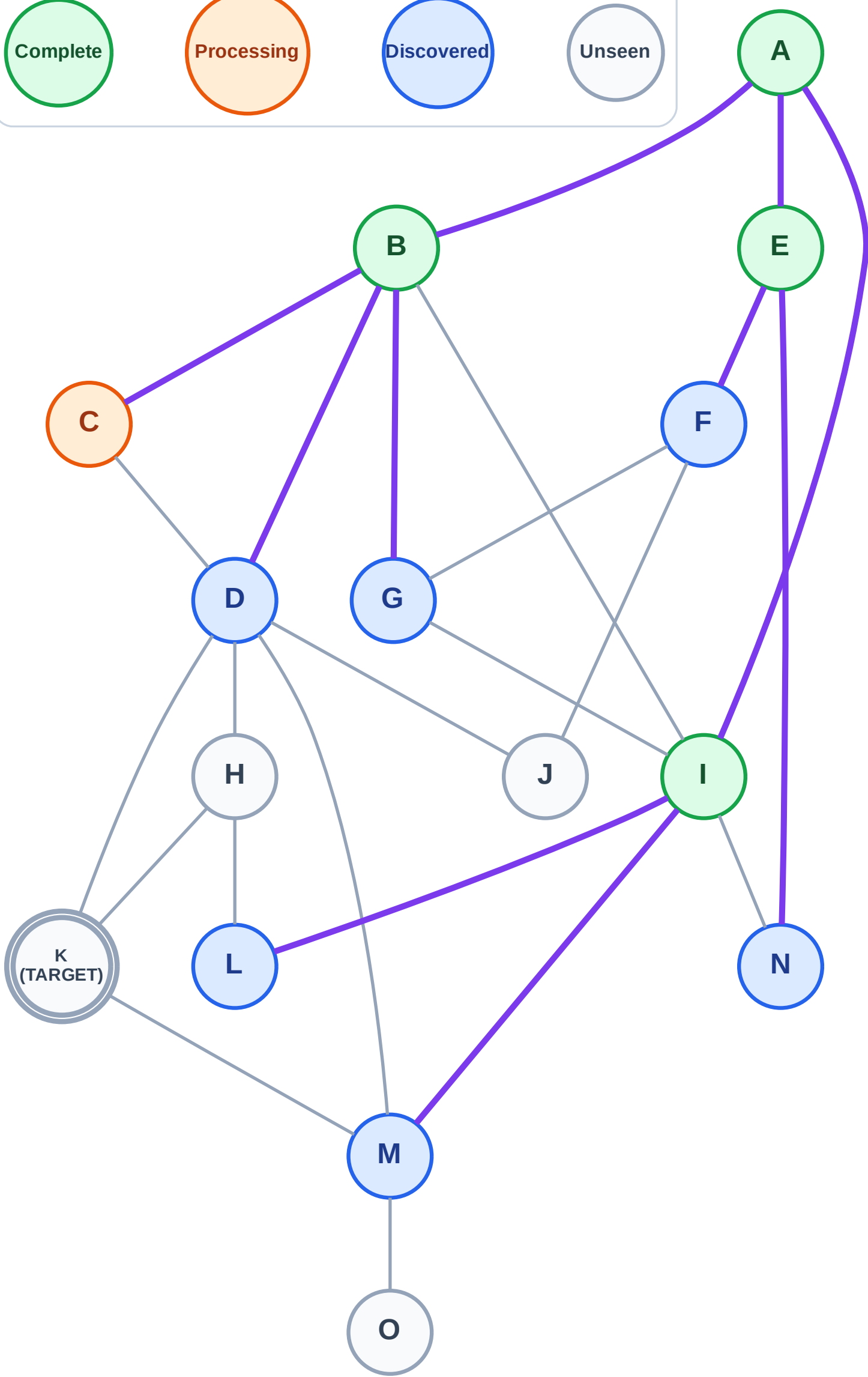
Processing

Discovered

Unseen

Queue: D → G → F → N → L → M

Visited: A, B, E, I



BFS Traversal

Step 11: No new neighbors from C

Legend

Complete

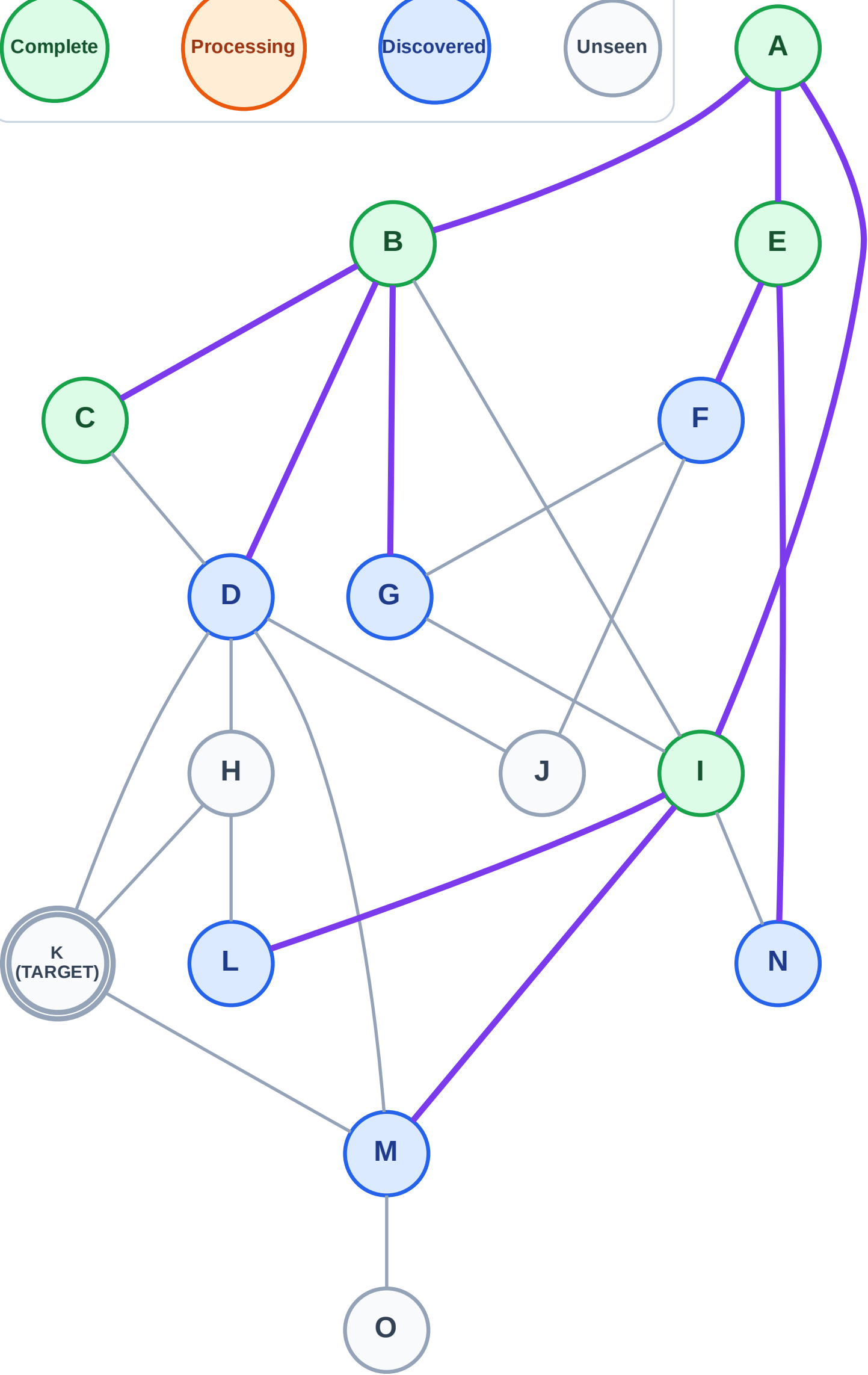
Processing

Discovered

Unseen

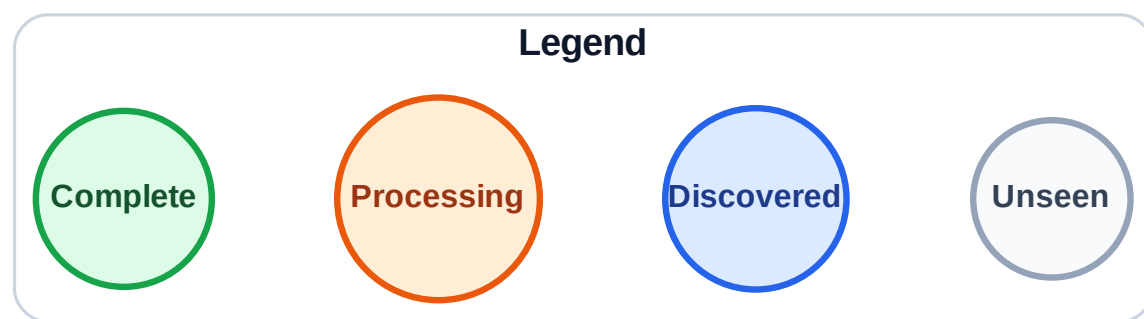
Queue: D → G → F → N → L → M

Visited: A, B, C, E, I



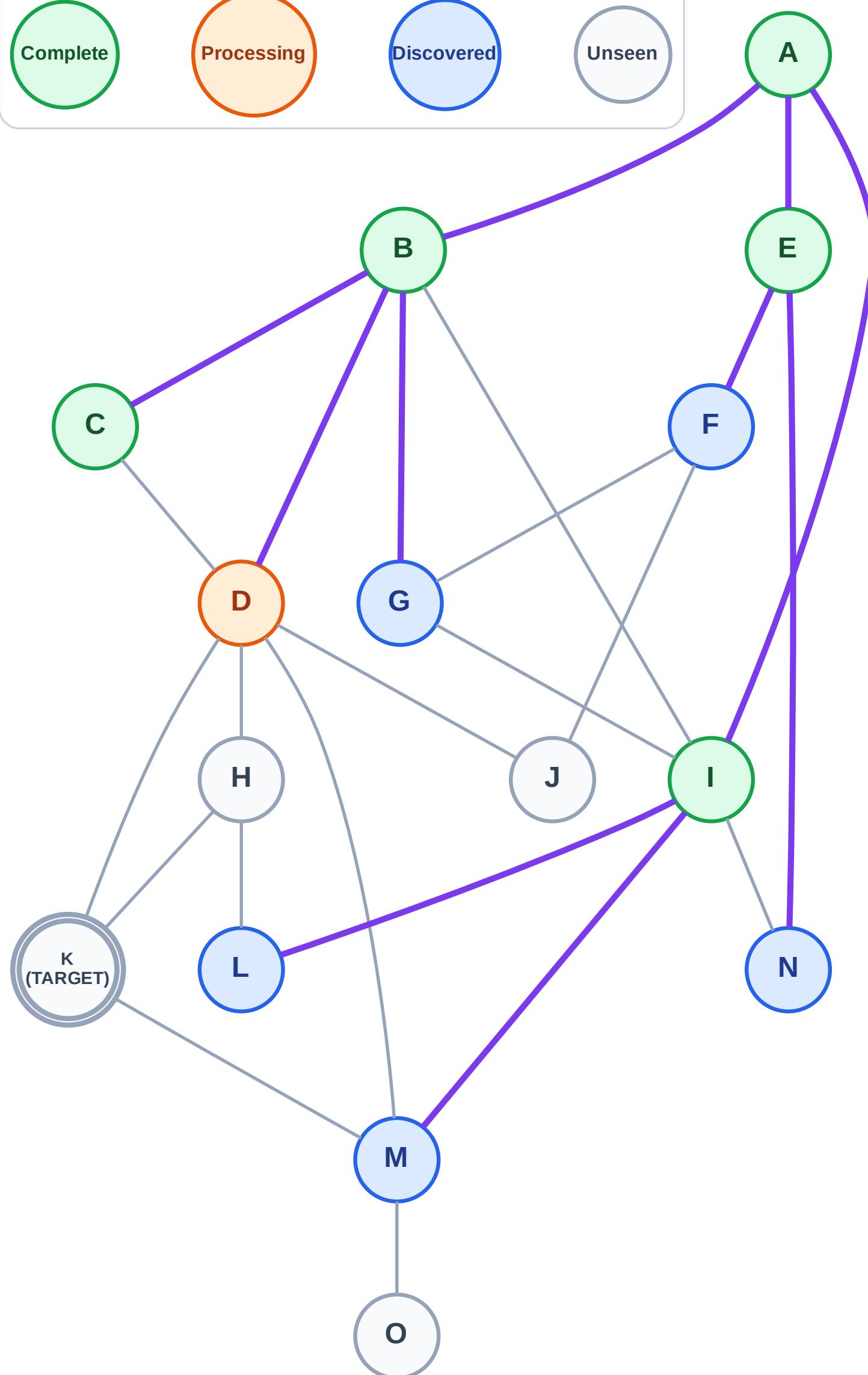
BFS Traversal

Step 12: Process node D



Queue: G → F → N → L → M

Visited: A, B, C, E, I



BFS Traversal

Step 13: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → D → K

Legend

Complete

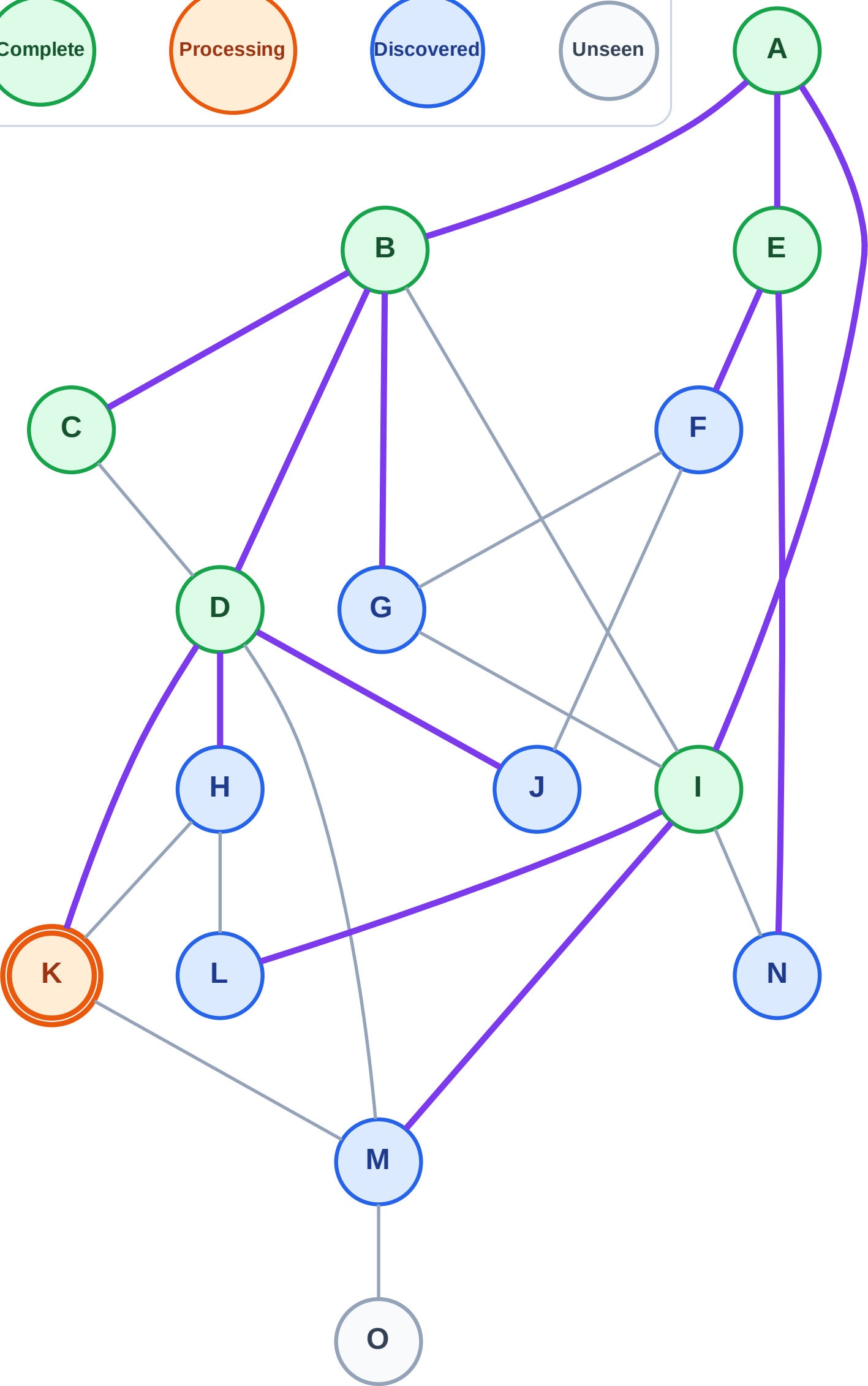
Processing

Discovered

Unseen

Queue: G → F → N → L → M → H → J

Visited: A, B, C, D, E, I, K



DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

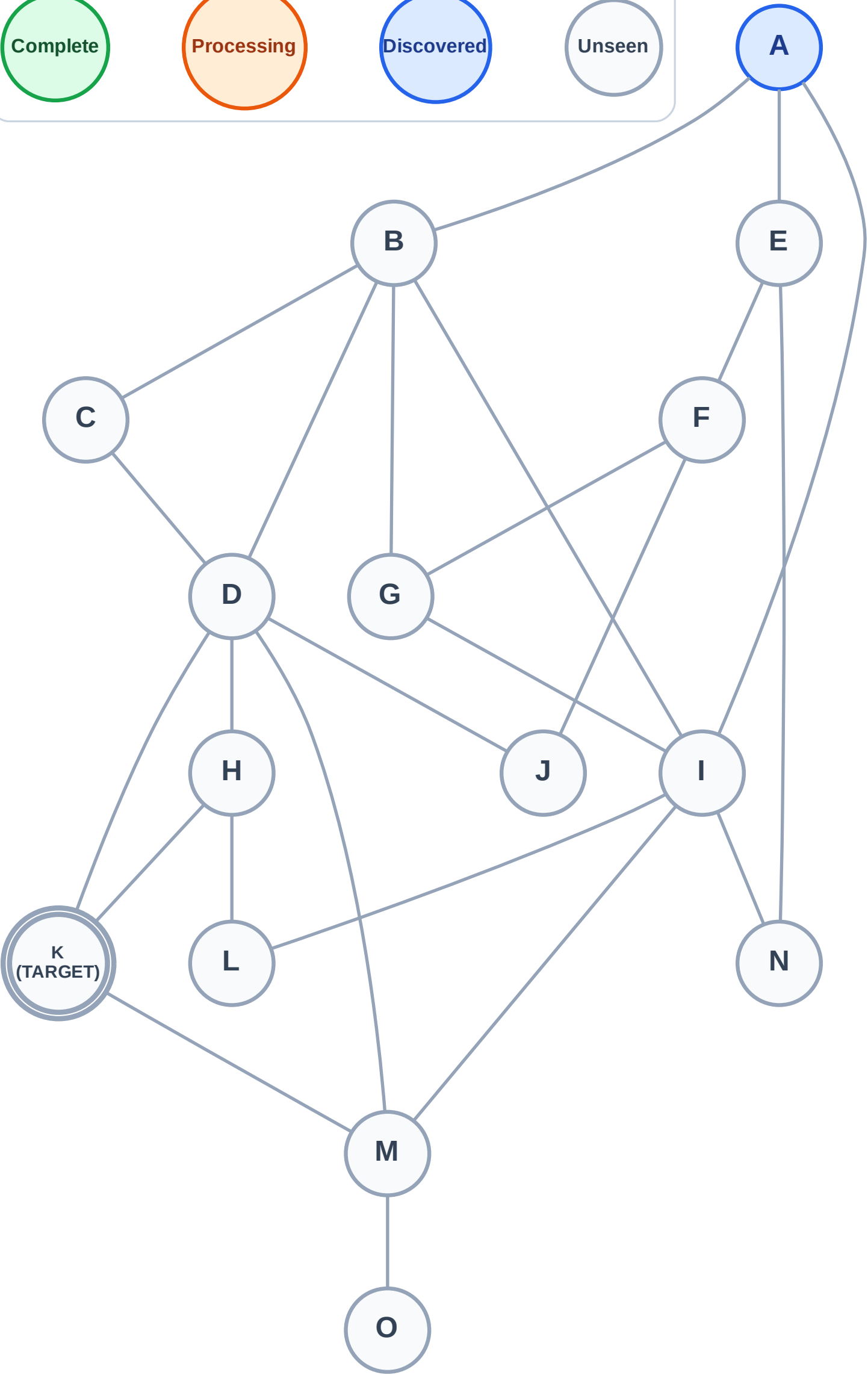
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

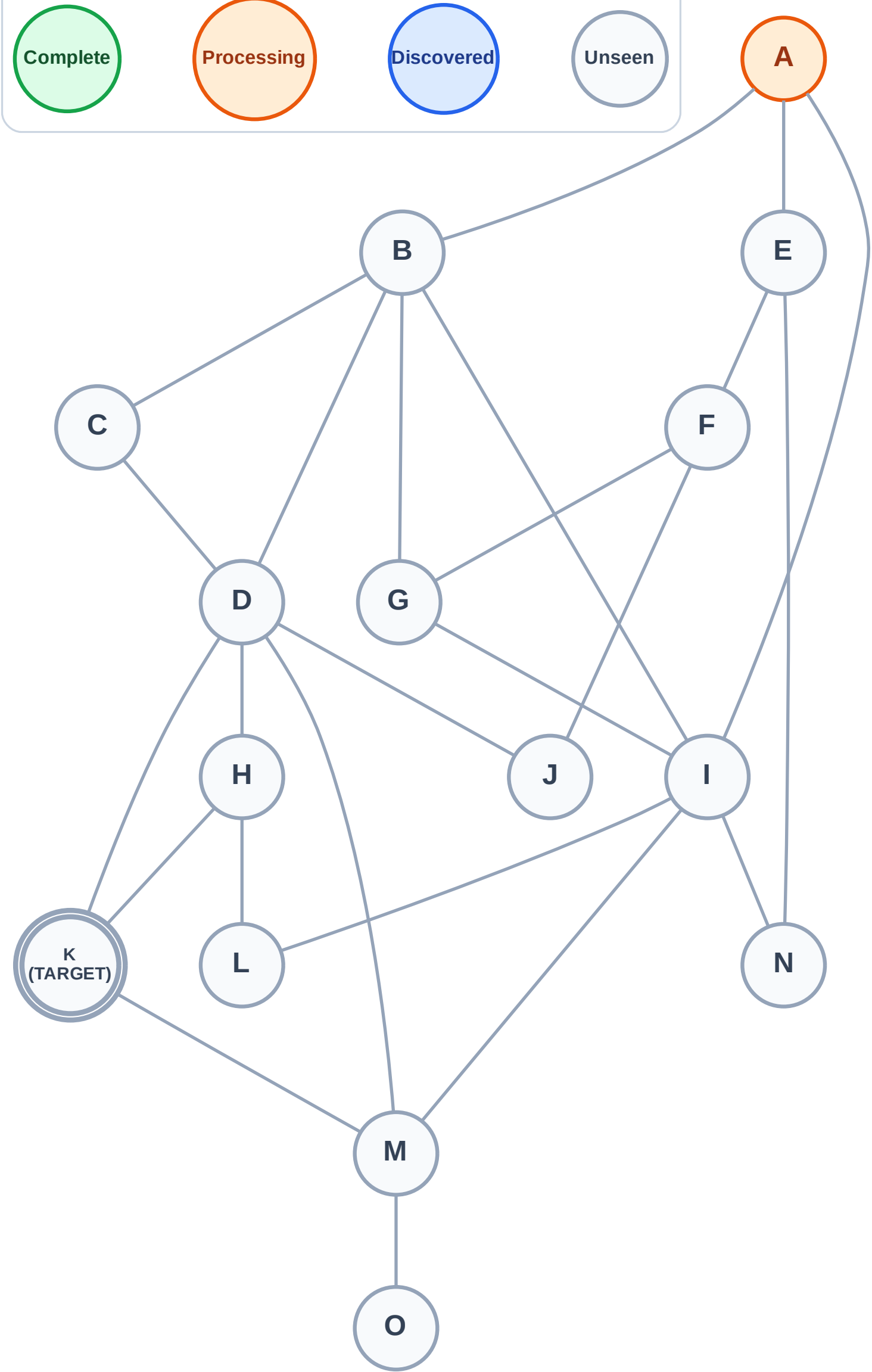
Complete

Processing

Discovered

Unseen

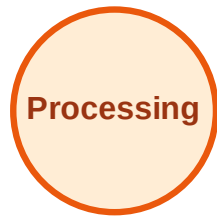
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

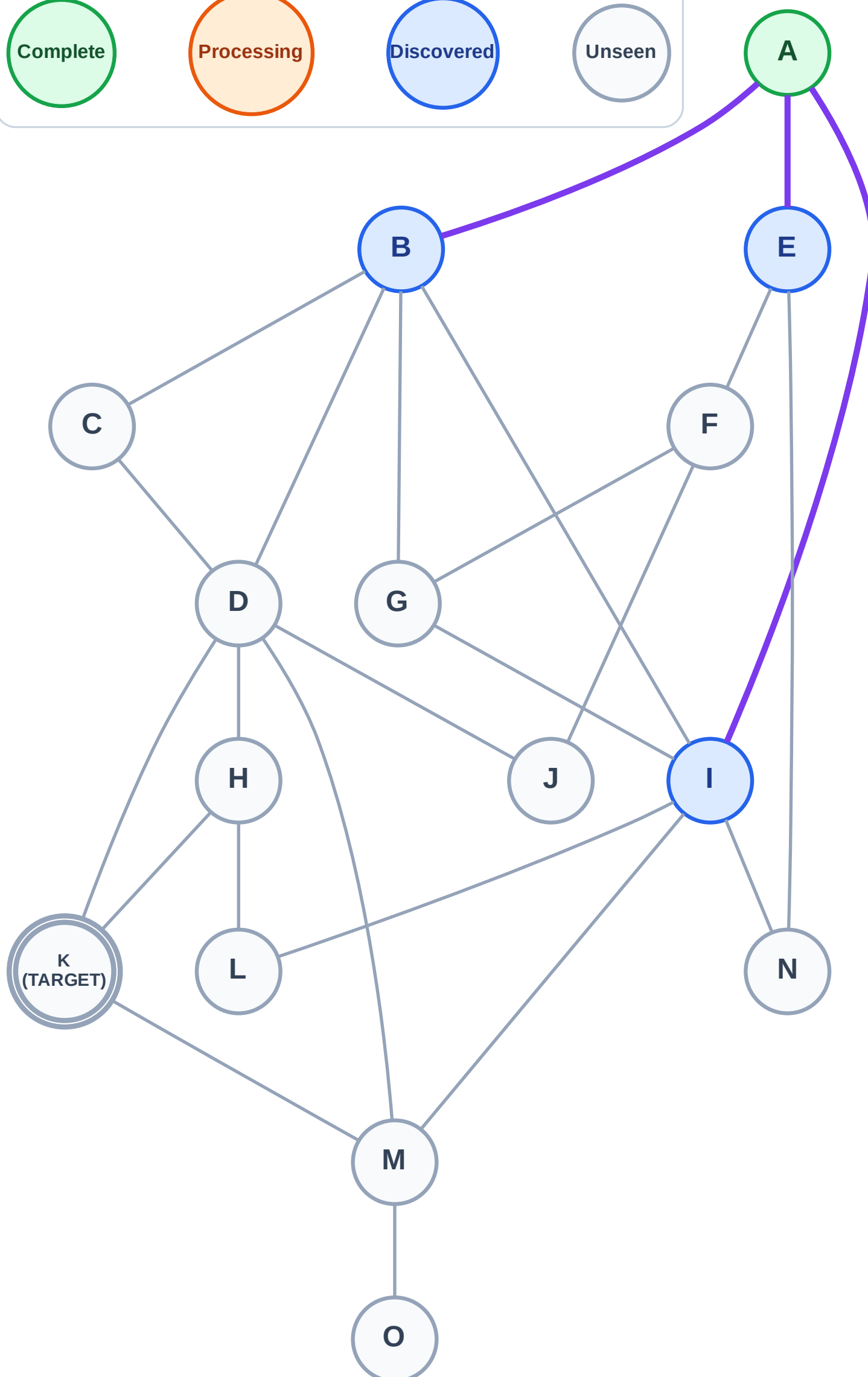
Step 3: Discover I, E, B from A

Legend



Stack (top at right): I | E | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

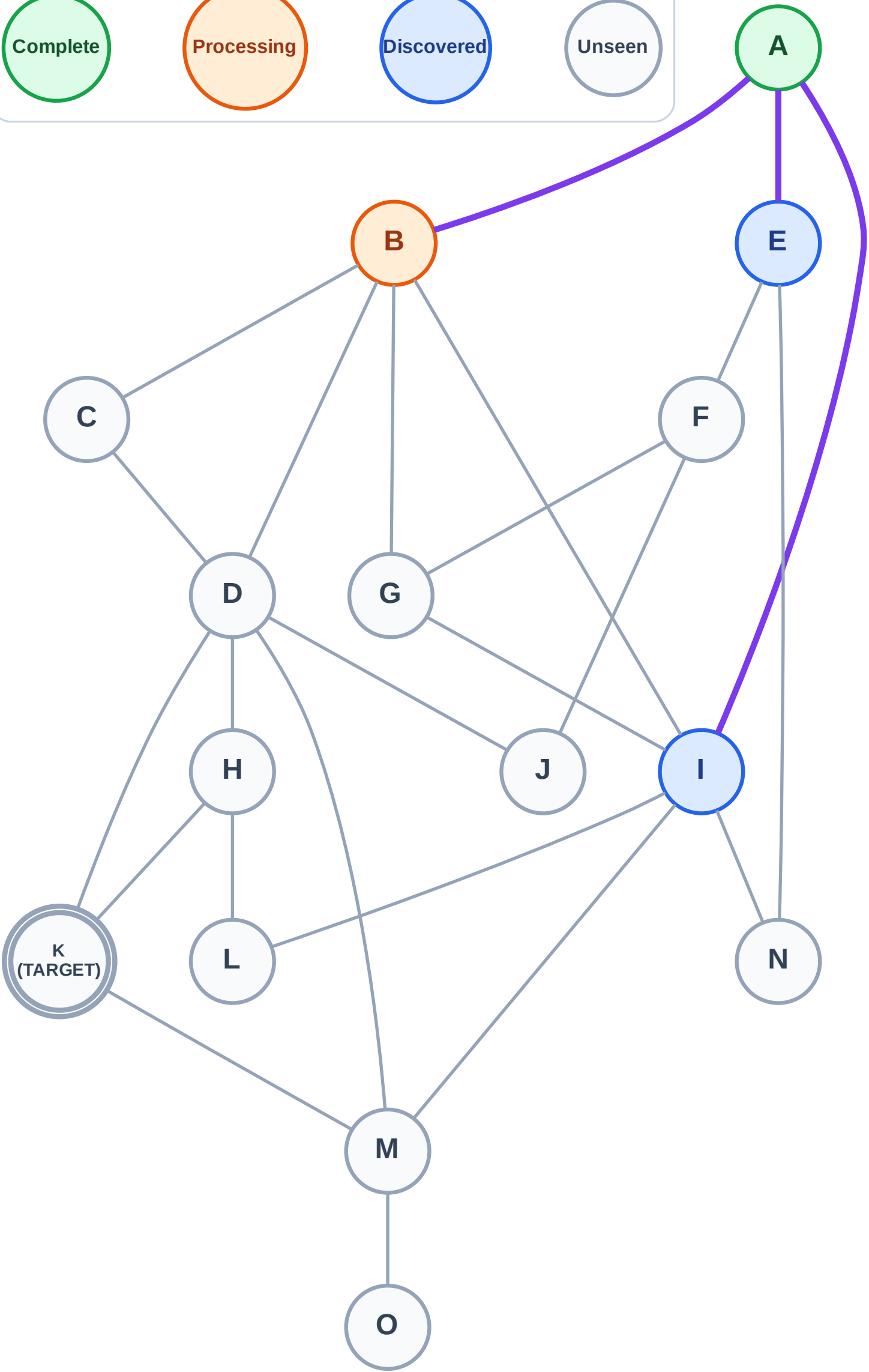
Processing

Discovered

Unseen

Stack (top at right): I | E

Visited: A



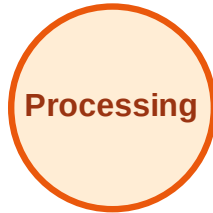
DFS Traversal

Step 5: Discover G, D, C from B

Legend



Complete



Processing



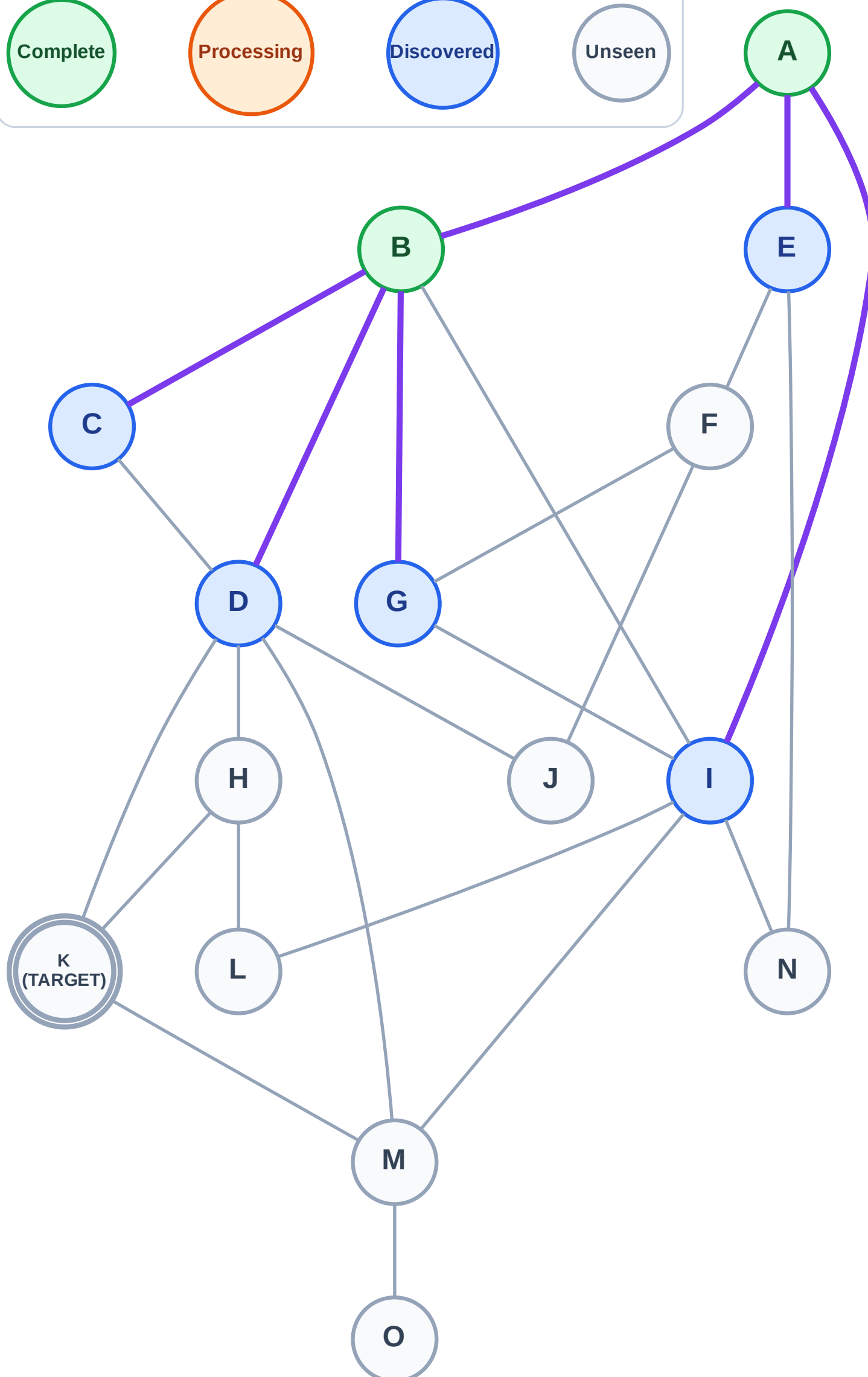
Discovered



Unseen

Stack (top at right): I | E | G | D | C

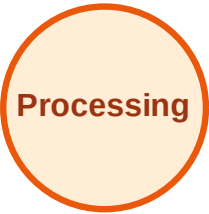
Visited: A, B



DFS Traversal

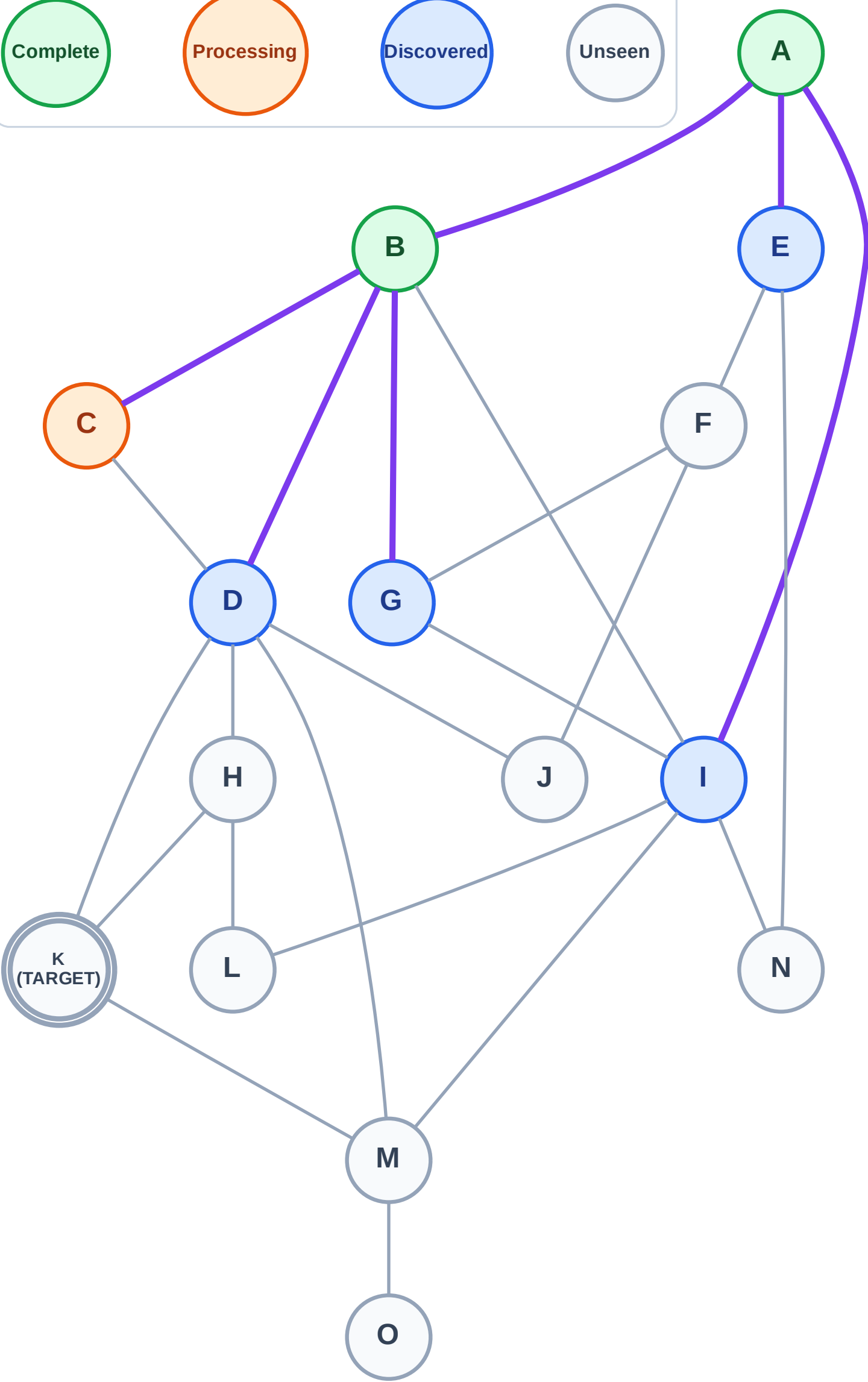
Step 6: Process node C

Legend



Stack (top at right): I | E | G | D

Visited: A, B



DFS Traversal

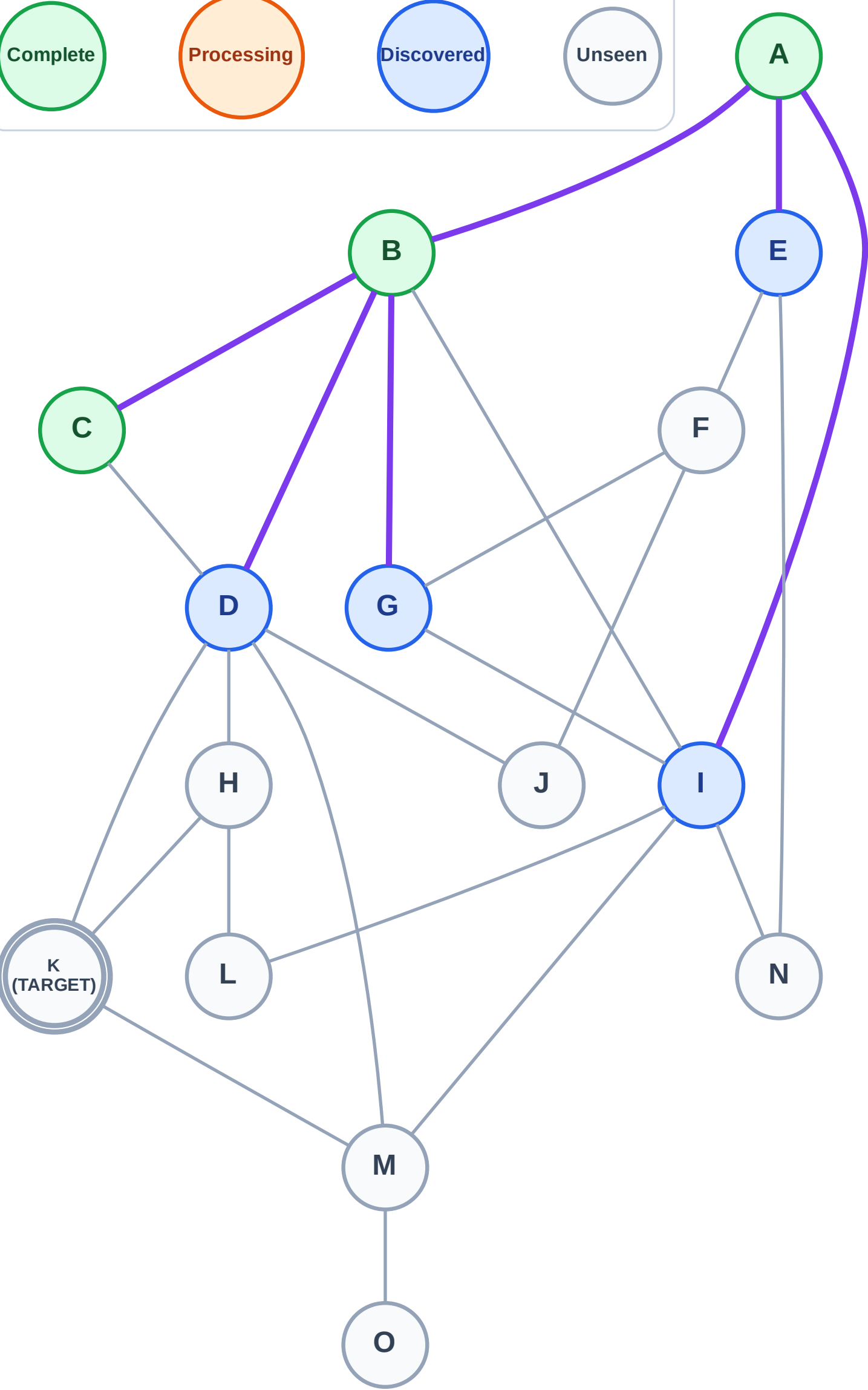
Step 7: No new neighbors from C

Legend



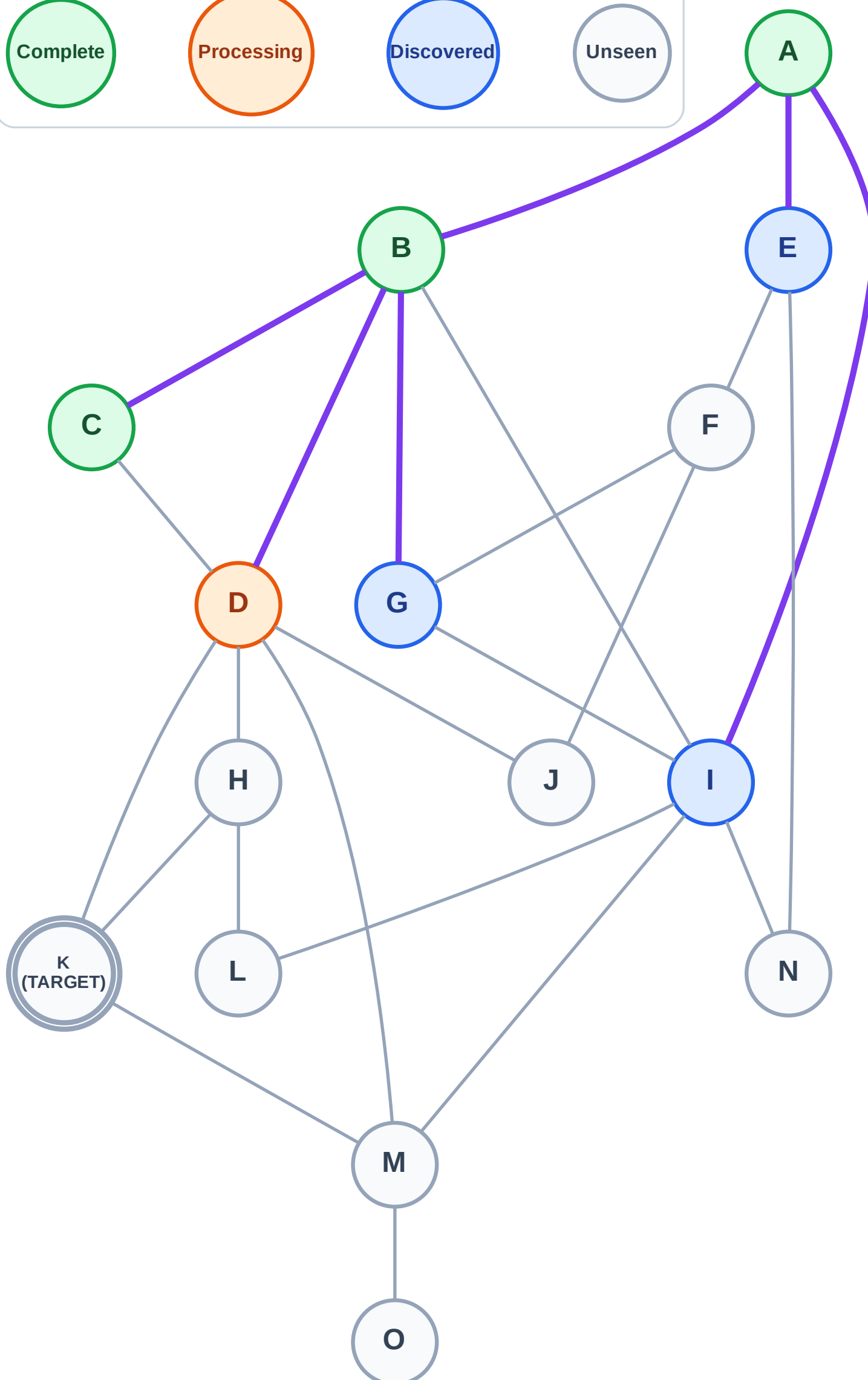
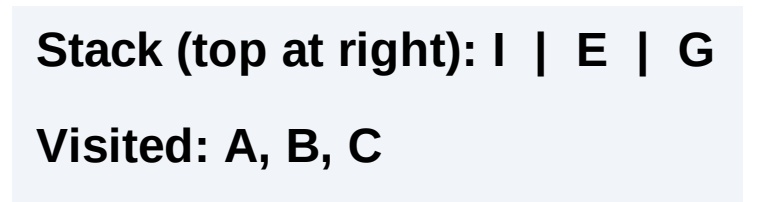
Stack (top at right): I | E | G | D

Visited: A, B, C



Step 8: Process node D

Step 8: Process node D



DFS Traversal

Step 9: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → D → K

Legend



Stack (top at right): I | E | G | M

Visited: A, B, C, D, K

